

What each of us owns, and the first thing to do

Team Vagabonds · Aavaran · problem statement SIH26171

This explains every person's part of the project, written so it makes sense with no technical background at all. For each person: what the thing actually is, what you own, the first concrete task, and what you can say when somebody asks.

Read your own section properly. Skim the others so you know who to point at when a question is not yours.

Why this matters more than it looks

We are asking the institute to nominate us. If that happens, at some point somebody senior asks one of us a question about our own area. The answer "I am not sure, that was someone else" is fine once. It is not fine from everybody.

So the point of this document is not to memorise a script. A script survives exactly one follow-up question. The point is that each of us owns something real, does the next piece of it this week, and can then talk about it because we actually did it.

Every task below is genuinely open work. None of it is invented to keep anyone busy. Several are listed in the report as things that still need doing, which means finishing them improves the submission, not just our story about it.

How the marks work

ISRO published exactly how entries are scored. Every task below connects to one of these.

What is being marked	Worth
Does the AI understand what is on the screen?	25%
Do we catch every private detail?	20%
Do we avoid covering up harmless things?	20%
Is the software light on your computer?	20%
How long does a real task take?	15%

Harsh Bajpai

Architecture and the redaction engine

What this is. The *architecture* is the decision about where each piece of the software lives, and in particular which piece is allowed to talk to the internet. The *redaction engine* is the part that finds every private detail on a page and covers it up before anything is sent.

What you own. The trust boundary, the engine that spots a PAN or an Aadhaar number and swaps it for a harmless label, and the loop that checks the AI's instruction is safe before carrying it out.

First task. Push the current code to the public repository. It is still at the 12 September version, so anyone cloning it today gets something that does not match the report. This blocks everything else.

If someone asks: "I worked on where the data is allowed to go. The part that talks to the internet never has the real values, so it cannot leak them even if we made a mistake in it."

Vansh Khosla

The repository and making it run everywhere

What this is. A *browser extension* is a small add-on for Chrome or Firefox, like an ad blocker. It has to be packaged in a way each browser accepts, and it has to work on computers that are not ours.

What you own. The repository itself, the builds for both browsers, and the automatic testing that runs on Windows, Mac and Linux every time the code changes.

First task. You own the repository on GitHub, so only you can move it to a team organisation. Once we agree a name, it is Settings, then Danger Zone, then Transfer ownership. Steps are in [ORG-MIGRATION.pdf](#). Right now the project URL in our report has one person's username in it, which undercuts a document about six people.

Second task. Run `./test-all.sh` on your own machine and tell us what happens. It has only ever been run on one laptop. Testing from a fresh clone has already caught two bugs that only appeared on a different computer, so this is not a formality.

If someone asks: "I look after the repository and the cross-browser builds, and the testing that proves it runs on Windows, Mac and Linux rather than just our own machines."

Manas Bharadia

Evaluation, the part that makes anyone believe us

What this is. *Evaluation* means testing the software and measuring how well it does, honestly. A *test page* is a realistic web page we test against. *Ground truth* is a hand-written answer key saying which details on that page are private and which are not, so the software can be marked against something that cannot argue back.

What you own. The test pages, their answer keys, and the honesty of every number in the report.

First task. Add real Indian government or bank pages to the test set. Save the page, then write an answer key listing which values are private and which only look private. This needs no coding: it is reading a page and filling in a list. Every single page added so far has exposed a real defect, which is why it is item 3 in the report's own plan.

Why it matters. This is what a professor attacks first. Anyone scores 100% on a test they wrote themselves. Our report says one score is 75% and explains why, and that honesty is what makes the other numbers believable. You own that.

If someone asks: "I work on testing and measurement. Some pages are written first and never tuned against, so the scores are honest rather than flattering."

Jinshri Jain

The Privacy Ledger, and how the project reads

What this is. The *Privacy Ledger* is a screen showing exactly what was covered up and exactly what was sent out. Two columns: what was on your screen, and what actually left your computer.

What you own. How that screen presents itself, and the visual language that carries into the deck and the report.

First task. Open Figure 2 on page 7 of the report and review it as a stranger would. It currently repeats three rows unnecessarily, the type is small, and it is not obvious at a glance which column is "before" and which is "after". Tell us what to change. You do not need to code it; describing the problem precisely is the work.

Why it matters. A privacy claim nobody can check is a slogan. That screen is what turns our claim into something a person verifies with their own eyes, and it only works if it is legible to someone who is not an engineer. It is the most persuasive picture in the document.

If someone asks: "I work on the screen that shows what was hidden and what was actually sent, so a user can check our privacy claim instead of taking our word for it."

Aarna Chauhan

The deck and the diagrams

What this is. The deck is the six slides that go to the SIH portal. The diagrams are the pictures that explain the system, including the one showing the dashed line with real data on one side and the internet on the other.

What you own. The submission deck, its figures, and how the report looks on the page.

First task. Slide 1 of the deck still has blanks: Theme, Team ID and Team Name. Theme is "Smart Automation", which is confirmed. The other two arrive when we register. Build the deck so those are the only things missing, and it can go out the moment we have them.

Second task. Read the report and tell us where it looks wrong. Not the words, the look. You have better judgement about that than the rest of us.

Why it matters. Our report admits a score of 75% that most teams would hide. That only works if the document looks like it came from people who knew what they were doing. A serious argument presented carelessly reads as a careless argument, and the person reading it is a researcher who judges documents for a living.

If someone asks: "I do the deck, the diagrams and how the report presents itself. The architecture diagram is the one picture that shows what the project actually is."

Siddhartha Chaudhary

The demonstration and the language review

What this is. The *demonstration* is the live run we show a judge. The *runbook* is the sheet saying exactly what to click, in what order, and what to do when something goes wrong. The *vocabulary lists* are the words the software uses to tell a private value from a harmless one, in each Indian language.

What you own. The demo, the runbook, and the language coverage.

First task. The software knows commercial words in eleven Indian languages, and eight of them have never been checked by somebody who reads the script. This matters concretely: a twelve-digit total on a bill passes the Aadhaar checksum, so if the word for "total" is missing in a language, the software covers up a harmless number and loses marks. The list is about five words per language. Find people who read Gujarati, Kannada, Malayalam, Punjabi, Odia, Marathi and Telugu, and get the words checked. The report lists this as work that "needs a reader rather than an engineer".

Second task. Run the demo end to end and time it. Write down what went wrong. That is the runbook.

If someone asks: "I work on the demonstration and on the Indian language coverage, because a word we are missing in Gujarati costs us marks on precision."

The honest bit

The split is not even, and this document is not pretending otherwise. Three of us are working mostly on code and three mostly on evaluation, presentation and demonstration.

Two things are true at once. The second half is what decides whether the first half convinces anybody: a correct system nobody can understand scores nothing. And the tasks above are genuinely open, so whoever does them has actually done them, and can say so without rehearsing anything.

"I worked on a different part of this, let me get you the right person" is a completely good answer to a question outside your area. It is a much better answer than guessing, and far better than an answer that falls apart on the second question.